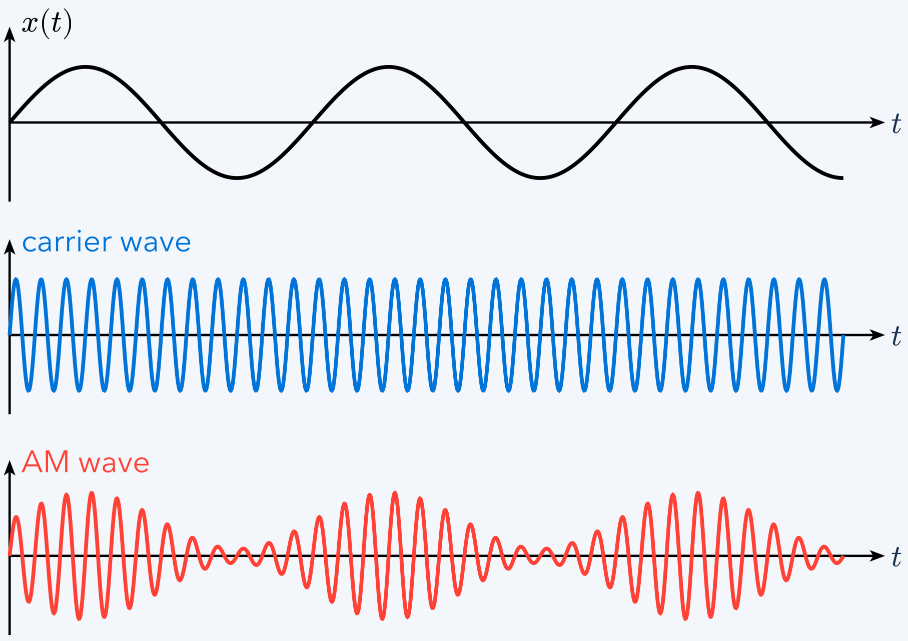
Amplitude vs Frequency Modulation

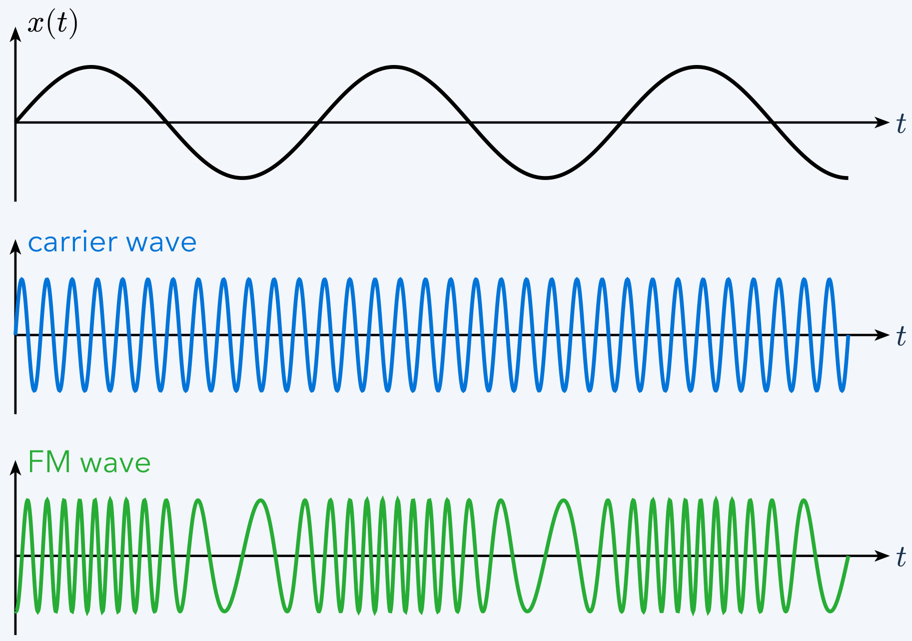
Transmit the same message by changing one property of a fast carrier. Read each column from the message at the top to the transmitted signal at the bottom.

AM: change the envelope



The carrier's amplitude follows the message; its rapid oscillation rate stays fixed. The envelope is easiest to see in the peaks.

FM: change the spacing



The instantaneous frequency follows the message; the amplitude stays fixed. Closely spaced peaks mean a higher frequency.

Memory aid: AM changes height; FM changes spacing. The carrier is the fast oscillation. The message is the slower signal we want to convey.